Explaining the Yang–Mills Mass Gap with Observer-Patch Repair Dynamics

A Support-Visible OPH Route to the Clay Problem: A Conditional Identity

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Abstract

The Clay Yang–Mills problem asks for a nontrivial four-dimensional quantum gauge theory with a positive mass gap. Observer-Patch Holography (OPH) supplies a finite repair mechanism whose relaxation gap equals the transfer-Hamiltonian gap when both operators come from the same ground-state transform. The four-dimensional conclusion also requires convergence of the Schwinger functions, reflection positivity, Euclidean locality, nontriviality, and convergence of the transfer data. These continuum and cross-fiber premises contain the hard physical construction. The paper states the resulting gap theorem conditionally and keeps the finite repair theorem separate from the Clay claim.

What This Paper Contributes

The standard target is clear: construct a nontrivial four-dimensional quantum Yang–Mills theory for every compact simple gauge group and prove a positive mass gap. OPH imports the usual compact gauge language, reflection positivity, and Osterwalder–Schrader/Wightman target conditions. The repair interpretation supplies a finite vacuum-sector relaxation mechanism.

On the support-visible compact-gauge branch, local holonomy data and the OPH scaling chart give the candidate Euclidean Yang–Mills form. Exact local repair acts as a positive Euclidean relaxation generator at finite cutoff. A uniform fixed-cutoff repair rate floor transports to the continuum Hamiltonian only when the transfer forms, vacuum projections, and finite intertwiners converge in the sense stated below. The proof therefore separates the finite repair mechanism from the additional four-dimensional continuum certificate required by the Clay problem.

1 Claim Boundary

This paper gives a proof-bearing projective weak-* / GNS extraction from the declared finite support-visible compact-gauge cylinder system. The finite positive-transfer and repair-gap mechanism is conditional on the finite ground-state-transform/cross-fiber and uniform-gap receipts. The finite ground-state-transform/cross-fiber receipt (Assumption 7.3) is the load-bearing unverified physical input of the paper: it assumes the gap-friendly spectral structure whose gap the later sections account for. Section 9 evaluates it on $\mathbb{Z}_2$ lattice gauge theory: the receipt holds analytically in the free limit, fails for the Wilson transfer matrix on the tested $L = 2, 3$ diagonal coupling grid, and gives the local Hamiltonian exactly in a fiber-dependent-rate form. The scan is not a no-go for other couplings, anisotropic limits, transfer objects, or that fiber-dependent-rate route. The Clay-facing Yang–Mills theorem additionally depends on the continuum certificate stated in Assumption 10.4: renormalized four-dimensional identification, reflection positivity and OS regularity, transfer/vacuum convergence, and noncollapse.

The finite local-domain source construction in the microphysics and particle papers carries a different exact positive gap [5, 6]. Its Hamiltonian is the sign-twisted graph Laplacian of one observer-visible seam complex under a declared reversing convention. It supplies neither the compact-gauge cylinder repair generator L_r^{rep}, the vacuum projector, the ground-state transform, a gauge-invariant excitation algebra, nor the uniform cofinal bound required here. That result therefore does not discharge the finite transfer receipt or any Clay-facing continuum premise.

More precisely, the imported four-dimensional form theorem uses compact-gauge reconstruction, the four-dimensional scaling chart, the reflection-positive ordinary vacuum sector, the absence of additional gauge-invariant relevant dimension-four pure-gauge operators beyond the positive quadratic curvature invariant, the support-visible cylinder extraction, and the renormalized regularity/universality receipt. The later mass-gap step adds the finite transfer receipt, exact atomic heat-bath collars, a finite source-type table, and a uniform L^2 approximate-tensorization/influence certificate.

Acceptance as a Clay-admissible solution depends on supplying the unproved parts of the OPH compact-gauge continuum certificate: renormalized Schwinger-function convergence, reflection positivity, Euclidean covariance/locality, nontriviality, and transfer/intertwiner convergence at the strength required by the Clay/Jaffe–Witten statement. The proof below then isolates the finite repair mechanism and shows that, on any branch carrying that certificate, the gap is identified exactly:

$$\Delta_{\mathrm{YM}} = \Delta_{\mathrm{rep}}.$$

The two-dimensional heat-kernel identity in the companion OPH results is a separate normalization and worldsheet-effective bridge; it is not the mass-gap proof.

2 Relation to the other OPH papers

This paper is a focused companion to the other OPH papers. The broad reconstruction framework is summarized in *From Observer Consensus to Standard Physics* [1]. The gauge derivation is *Deriving Standard Model Gauge Structure from Observer Overlap Consistency* [2], which carries the support-visible compact-gauge repair-gap theorem inside the Standard Model gauge paper itself. The particle branch is separate [3]. The finite repair and quotient-normal-form machinery comes from *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics* [4]. The regulated screen, record, and edge heat-kernel architecture is developed in *Federated Echosahedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in OPH* [5].

The companion edge-sector theorem relates OPH heat-kernel weights to a two-dimensional Yang–Mills partition identity. That identity fixes normalization and the controlled worldsheet effective bridge. The four-dimensional mass-gap argument uses repair dynamics: exact local repair becomes a positive Euclidean relaxation generator, and the uniform repair gap is transported to the compact-gauge Hamiltonian only under the continuum certificate.

The repair mechanism keeps records readable under continued reading at finite cutoff. The mass gap is then the cost of leaving the repair-fixed vacuum sector: the same consistency requirement that forces records at the substrate level supplies the relaxation generator whose spectral floor is Δ_{rep}. This interpretation adds no strength to the Clay-facing claim: the identity $\Delta_{\mathrm{YM}} = \Delta_{\mathrm{rep}}$ is conditional on (Y1)–(Y6) and the continuum certificate exactly as stated above.

The Clay-facing import is The Standard Model gauge paper's Yang–Mills theorem surface: the compact-gauge reconstruction ladder, the four-dimensional Euclidean Yang–Mills form theorem, the coherent compact-gauge extraction proposition, the conditional support-visible Osterwalder–Schrader reconstruction theorem, the nontriviality certificate, and the exact finite repair-gap theorem. This note isolates that branch theorem surface; it does not enlarge it.

3 The Clay Target

The Clay Mathematics Institute describes the Yang–Mills mass gap as the missing mathematical foundation behind the quantum theory used for nonabelian gauge forces [10]. Jaffe and Witten state the problem as follows: for every compact simple gauge group G, construct a nontrivial quantum Yang–Mills theory on $\mathbb{R}^4$ satisfying axiomatic properties at least as strong as the stated Wightman or Osterwalder–Schrader references and prove a mass gap $\Delta > 0$ [11, 12].

The OPH proof below addresses that target through a different variable. The starting data are finite observer patches, compact-gauge visible quotient data, and exact repair collars. The nonzero energy threshold is the cost of leaving the repair-fixed vacuum sector.

4 Standing Setup

Fix a compact simple gauge group G carried by an OPH compact-gauge zero-obstruction vacuum branch, realized at fixed cutoff by the declared compact-gauge patch-carrier architecture. The architecture supplies finite local Hilbert spaces, exact local constraints, patch and overlap algebras, overlap sector projectors, record layers, and local repair interfaces.

Let r range over a cofinal refinement family of finite regulators. For each r, let

$$(\mathcal{H}_r, \Omega_r, H_r), \qquad T_r(t) = e^{-tH_r},$$

be the physical Euclidean Hilbert space, vacuum, Hamiltonian, and transfer semigroup.

Let $X_r := X_r^{(0)}$ be the support-visible time-zero compact-gauge quotient configuration space, let π_r be the stationary time-zero measure, and set

$$K_r := L^2(X_r, \pi_r).$$

Write $\omega_r(a) := \int_{X_r} a \, d\pi_r$ for the associated state functional; K_r implements the finite π_r-null support reduction by definition. Let $\mathcal{C}_r$ be the finite family of active repair collars. For each active collar $C \in \mathcal{C}_r$, let

$$\rho_C := \rho_{C,r} : X_r \to Y_{C,r}$$

be the complete repaired visible datum, and let

$$E_C : K_r \to K_r$$

be conditional expectation onto the ρ_C-measurable functions.

The proof stays on the ordinary or central zero-obstruction vacuum branch. The genuinely noncentral higher-gauge branch is a different fixed-cutoff sector in the OPH framework and is not used for the ordinary compact-simple G theorem below.

Notation	Meaning
X_r, π_r, K_r	support-visible time-zero quotient configuration space, stationary measure, and $L^2(X_r, \pi_r)$.
$\mathcal{C}_r$	finite active repair-collar family at regulator r.
ρ_C	complete repaired visible datum on collar C.
E_C	π_r-preserving conditional expectation onto ρ_C-measurable functions.
L_r^{rep}	ground-state transformed Euclidean repair generator at cutoff r.
$P_{0,r}$	projection onto constants in K_r, corresponding to the physical vacuum.
U_r	finite-stage unitary from the physical Euclidean Hilbert space to the repair L^2 space.
$K, \mathcal{H}, U$	support-visible continuum repair Hilbert space, physical Hilbert space, and limiting unitary.
c_*	uniform active-collar repair-rate floor.
A_*, δ_*	uniform approximate-tensorization constant and certified global gap c_*/A_*.

5 Imported 4D Euclidean Yang–Mills Form

Assumption 5.1 (Support-visible compact-gauge Yang–Mills branch)**.** *The compact-gauge branch used below satisfies the following branch-local conditions.*

(i) the ordinary or central zero-obstruction compact-gauge sector survives refinement with compact simple structure group G;

(ii) the cofinal regulator tail satisfies The Standard Model gauge paper's explicit compact-gauge refinement receipt;

(iii) the support-visible quotient carries a four-dimensional Euclidean scaling chart;

(iv) the ordinary vacuum sector is reflection positive and has topological angle $\theta = 0$;

(v) the local finite-constraint MaxEnt/Gibbs family is gauge-invariant, Euclidean local, rotation-invariant, and refinement-stable;

(vi) no additional gauge-invariant relevant dimension-four pure-gauge operator remains on this branch besides the positive quadratic curvature invariant;

(vii) the branch is carried on a separated cofinal regulator system with the finite compact-gauge cylinder data defined in Section 10.

No continuum cylinder state, GNS space, transfer generator, or vacuum projection is assumed in this branch declaration.

Theorem 5.2 (Imported conditional OPH four-dimensional Euclidean Yang–Mills form)**.** *Under Assumptions 5.1 and 10.4, on an extracted cylinder family from Proposition 10.2 that lies on the declared local four-dimensional scaling branch, the continuum gauge-sector Euclidean action is*

$$S_E[A] = \frac{1}{4g^2}\int_{\mathbb{R}^4} \langle F_{\mu\nu}, F_{\mu\nu}\rangle \, d^4x, \qquad F = dA + A \wedge A, \tag{YM}$$

with compact simple structure group G. The extracted gauge-quotient cylinder family is denoted

$$d\mu_{\mathrm{YM}}(A) = Z^{-1} e^{-S_E[A]} \, DA/G$$

in the OPH support-visible GNS representation. This notation abbreviates the certified cylinder family; it is not a literal infinite-dimensional Lebesgue measure. Its support-visible continuum transfer semigroup is the corresponding Euclidean Yang–Mills semigroup.

Proof. This is the support-visible compact-gauge Yang–Mills form theorem of the compact OPH paper, where it appears as Theorem `thm:oph-4d-euclidean-yang-mills-form` [2]. The proof spine is recalled here because it fixes the target Hamiltonian for the mass-gap step.

The present Yang–Mills branch takes compact simple G as branch data. When G is imported from OPH reconstruction, that import is conditional on the compact-gauge refinement receipt; on that tail the zero-obstruction transportable bosonic sector category and its faithful forgetful fiber functor reconstruct G [2]. At fixed cutoff, the declared compact-gauge patch-carrier presentation gives support-visible link holonomies and plaquette holonomies. In the refinement limit, the zero-obstruction gluing law makes infinitesimal rectangle holonomies multiplicative and path-local. The Cauchy/regularity certificate in items (Y1)–(Y3) upgrades the generalized-holonomy cluster state to a local connection A on the four-dimensional scaling chart, and the infinitesimal plaquette defect is

$$U_{\mu\nu}(\varepsilon, x) = \mathbf{1} + \varepsilon^2 F_{\mu\nu}(x) + O(\varepsilon^3), \qquad F = dA + A \wedge A.$$

The Euclideanized MaxEnt/local-Gibbs branch gives a local finite-range action density built from support-visible gauge-invariant collar data. Gauge quotienting permits only class functions of the curvature and its covariant derivatives. The four-dimensional scaling chart, Euclidean rotation invariance, locality, and reflection positivity leave one relevant dimension-four positive quadratic invariant in the pure gauge sector:

$$\langle F_{\mu\nu}, F_{\mu\nu}\rangle.$$

The possible topological density $\langle F \wedge F\rangle$ is reflection odd and belongs to a separate topological-angle sector; it is absent on the ordinary reflection-positive zero-obstruction vacuum branch used here. Higher curvature powers and covariant-derivative terms are irrelevant operators under the declared continuum scaling; their disappearance in the extracted state is part of the uniform renormalized remainder control in (Y1)–(Y3). Normalizing the unique positive quadratic invariant defines the coupling g and gives (YM).

The finite-stage cylinder measures are the gauge-register / quantum-link Gibbs measures pushed to the support-visible quotient. Proposition 10.2 proves the compatible support-visible weak-$*$ / GNS cylinder extraction. The generator and transfer-semigroup identification use the separate dynamic items (Y4)–(Y5) of Assumption 10.4; weak-* compactness alone does not provide them. ☐

Corollary 5.3 (Yang–Mills equations on the compact-gauge branch)**.** *With an external conserved gauge current J, the OPH curvature-square action on the same support-visible compact-gauge branch gives*

$$DF = 0, \qquad D{*}F = g^2{*}J,$$

equivalently

$$D_\mu F^{\mu\nu} = g^2 J^\nu$$

in local coordinates after the usual continuation to the Lorentzian field-equation convention.

Proof. The curvature $F = dA + A \wedge A$ obeys the covariant Bianchi identity $DF = 0$. Varying the quadratic curvature action

$$S[A, J] = -\frac{1}{2g^2}\int \langle F \wedge {*}F\rangle + \int \langle A \wedge {*}J\rangle$$

with respect to compactly supported variations of A gives

$$\delta S = \int \langle \delta A \wedge (g^{-2} D{*}F - {*}J)\rangle$$

up to a boundary term. Stationarity gives $D{*}F = g^2{*}J$. The source-free equation is the case $J = 0$. ☐

Remark 5.4 (Why this step matters for the prize)*.* The proof has two separate claims. Theorem 5.2 identifies the support-visible continuum gauge sector with four-dimensional Euclidean Yang–Mills under its branch hypotheses. The spectral argument applies to that Hamiltonian and proves a positive gap only when Assumption 10.4 supplies the OS/transfer/nontriviality bridge.

Remark 5.5 (Abelian boundary and Maxwell branch)*.* The ordinary electromagnetic $\mathrm{U}(1)_Q$ branch is the abelian boundary case of the same compact-gauge curvature package. Restricting $F = dA + A \wedge A$ to $\mathfrak{u}(1)_Q$ gives

$$F_Q = dA_Q, \qquad dF_Q = 0, \qquad d{*}F_Q = g_Q^2{*}J_Q$$

after varying the quadratic electromagnetic action with current J_Q. This recovers Maxwell's equations after canonical electromagnetic normalization. The Clay-facing mass-gap claim concerns compact simple nonabelian G on the support-visible branch above. On the separate abelian branch, the displayed Maxwell kinetic action, an ordinary Lorentz vacuum, and the absence of a Higgs, Stückelberg, or medium mass give two transverse classical modes with quadratic pole $k^2 = 0$. The abstract group $\mathrm{U}(1)_Q$ alone does not imply that action, phase, or pole. A photon particle additionally requires a positive-energy physical quantization and a positive-residue massless spectral pole; none of those quantum receipts is part of the Clay-facing Yang–Mills gap statement.

6 Fixed-Cutoff Repair Equals Projection

Proposition 6.1 (Local exact repair equals conditional expectation)**.** *For each active collar C, the exact-Markov repair map on the support-visible quotient is the π_r-preserving conditional expectation E_C.*

Proof. On the exact-Markov branch, repair preserves exactly the repaired visible datum ρ_C, changes only complementary invisible fiber data, and acts on the quotient-first physical algebra rather than on microscopic representatives. Let Φ_C be the Heisenberg repair map and let $\mathcal{N}_C$ be the repaired local fixed algebra. Exact repair semantics require $\mathcal{N}_C$ to be exactly the ρ_C-measurable subalgebra. They give

$$\Phi_C(a) = a \quad (a \in \mathcal{N}_C),$$

$$\Phi_C(\mathcal{A}_r^{G,\mathrm{sv}}) \subseteq \mathcal{N}_C, \qquad \omega_r \circ \Phi_C = \omega_r,$$

and Φ_C is $\mathcal{N}_C$-bimodular. Therefore, for $a \in \mathcal{N}_C$ and $x \in \mathcal{A}_r^{G,\mathrm{sv}}$,

$$\omega_r(a^* \Phi_C(x)) = \omega_r(a^* x).$$

Since $\Phi_C(x) \in \mathcal{N}_C$, this equation characterizes its class in K_r as the orthogonal projection of x onto the ρ_C-measurable subspace, uniquely modulo π_r-null functions. That orthogonal projection is the π_r-preserving conditional expectation E_C. □

7 Exact Euclidean-Consensus Law

Lemma 7.1 (Fiber-homogeneous orbit condition)**.** *Fix an active collar C and a repaired value $y \in Y_{C,r}$. On the support-visible quotient, let the complete conditional fiber be*

$$F_C(y) = \rho_{C,r}^{-1}(y)$$

with conditional law $\nu_{C,y}$. If this is either finite uniform or standard atomless, has no remaining observable labels, and the primitive relaxation is invariant under every $\nu_{C,y}$-preserving automorphism, then the full fiber, including every holonomy coordinate actually resampled by repair, is homogeneous for the local repair receipt.

Proof. This is the complete-fiber homogeneity receipt. Quotienting removes declared implementation labels, but the proof must also check that no remaining source constraint or relaxation rate distinguishes points in the fiber. Under that check, the conditioned state and primitive relaxation carry the stated full measure-preserving symmetry. □

Lemma 7.2 (Scalar relaxation on a homogeneous conditional fiber)**.** *Let (F, ν) be either a finite uniform probability space or a standard atomless probability space. Let E_F be expectation onto constants, and let D_F be a bounded positive self-adjoint Markov relaxation generator such that*

$$\ker D_F = \mathrm{Ran}(E_F)$$

and D_F commutes with every measure-preserving automorphism of (F, ν). Then there is a scalar $c_F > 0$ such that

$$D_F = c_F(I - E_F).$$

Proof. For a finite uniform fiber, the symmetric-group representation on the mean-zero subspace is irreducible. For a standard atomless fiber, use equal-measure finite partitions. Automorphisms within pieces and permutations of pieces force every bounded commutant operator to be scalar on the mean-zero step functions; compatibility under refinements gives one scalar, and these step functions are dense in $L^2_0(F, \nu)$. Positivity and the kernel condition make the scalar strictly positive. The formula follows. □

Assumption 7.3 (Finite ground-state-transform and cross-fiber receipt)**.** *For every regulator, the reflection-positive finite transfer matrix supplies a unitary $U_r : \mathcal{H}_r \to K_r$, with $U_r\Omega_r = \mathbf{1}_r$, such that*

$$U_r H_r U_r^{-1} = \sum_{C\in\mathcal{C}_r} D_C.$$

Each D_C is a bounded positive self-adjoint detailed-balance Markov generator acting on the complete complementary conditional fiber, including every record or holonomy coordinate changed by repair; its fixed algebra is exactly the $\rho_{C,r}$-measurable algebra, and its fiberwise scalar $c_C(y)$ is independent of the repaired value y on the support of π_r. A Gibbs state by itself does not imply this finite transfer-matrix/decomposition receipt.

Theorem 7.4 (Exact Euclidean repair law under homogeneous fibers)**.** *Under the fiber-homogeneous orbit condition for every active collar and Assumption 7.3, there are positive constants $c_C > 0$ such that the ground-state transformed physical Euclidean generator is exactly*

$$L_r^{\mathrm{rep}} = \sum_{C\in\mathcal{C}_r} c_C(I - E_C), \tag{1}$$

and therefore

$$U_r e^{-tH_r} U_r^{-1} = e^{-tL_r^{\mathrm{rep}}} \qquad (t \geq 0), \tag{2}$$

for a unitary $U_r : \mathcal{H}_r \to K_r$ with $U_r\Omega_r = \mathbf{1}_r$.

Proof. The finite receipt supplies the ground-state transform and its primitive local pieces. Each D_C is supported on one collar C, preserves exactly the repaired visible datum $\rho_{C,r}$, and relaxes the complete complementary conditional fiber. Hence

$$\ker D_C = \mathrm{Ran}(E_C).$$

Lemma 7.1 gives the full hidden-fiber permutation symmetry. Applying Lemma 7.2 fiberwise gives $D_{C,y} = c_C(y)(I - E_{C,y})$. The cross-fiber receipt makes $c_C(y) = c_C$, so

$$D_C = c_C(I - E_C).$$

The scalar is positive because C is active. Branch homogeneity makes it a collar-type scalar, so summing over the active collars gives (1), and exponentiation of the positive self-adjoint generator gives (2). □

Definition 7.5 (Source-defined admissible atomic collar tower)**.** *For every regulator and allowed boundary condition b, let the source give an atomic register set V_r with finite local alphabets, a finite-range Gibbs law $\pi_{r,b}$, and one rooted active collar $C(v)$ per $v \in V_r$, with*

$$P_{v,r,b}f := \mathbb{E}_{\pi_{r,b}}[f \mid x_{V_r\setminus\{v\}}].$$

Its source type is the isomorphism class of the rooted interaction-radius neighbourhood, boundary/sector flags, local alphabets, nonzero potential templates, complete repaired readback, conditional kernel, and normalized rate. The tower is admissible when:

(G1) *all active types over every location, allowed boundary, system size, and cofinal refinement stage belong to one printed finite set $\mathfrak{T}_{\mathrm{act}}$;*

(G2) $\mathcal{C}_r = \{C(v) : v \in V_r\}$, $E_{C(v)} = P_{v,r,b}$, *and type-equivalent collars have the same rate* $c_{v,r,b} = c_{\tau(v)} \geq c_* > 0$*;*

(G3) *if* $a_{vu}^{r,b}$ *is the supremum total-variation change of the root-*v *conditional kernel when exterior configurations differ only at* u*, outward-rounded rational bounds from the finite type table prove*

$$\sup_{r,b,v} \sum_{u \neq v} a_{vu}^{r,b} \leq \eta_* < 1. \tag{3}$$

The Dobrushin Poincaré comparison then derives the uniform approximate-tensorization bound with $A_* = (1-\eta_*)^{-1}$ *[17];*

(G4) *coarse shadow preserves the listed type transitions and conditional kernels and creates no unlisted type.*

Qualitative finite-range mixing and collar-CMI decay are not item (G3)*.*

Proposition 7.6 (Finite collar classification and local-rate floor)**.** *The admissible collar classes are exactly the nonempty fibers of* $\tau : \bigsqcup_r V_r \to \mathfrak{T}_{\mathrm{act}}$*. Hence there are at most* $|\mathfrak{T}_{\mathrm{act}}|$ *classes and*

$$c_{v,r,b} \geq c_* := \min_{t \in \mathfrak{T}_{\mathrm{act}}} c_t > 0 \tag{4}$$

uniformly across location, boundary condition, system size, and cofinal refinement.

Proof. The type tuple is a complete source signature. Items (G1)–(G2) therefore identify precisely a finite image with a positive finite minimum, while item (G4) prevents refinement from producing a new or rate-degenerate type. □

8 Finite-Stage Gap

Theorem 8.1 (Uniform collar-projection and transfer gap)**.** *For an admissible collar tower satisfying the finite ground-state-transform receipt, set*

$$\delta_* := c_*(1-\eta_*) = \frac{c_*}{A_*} > 0 \tag{5}$$

in the influence-certified case. Uniformly across the complete family,

$$L_{r,b}^{\mathrm{rep}} \geq \delta_*(I - P_{0,r,b}), \qquad \|e^{-tL_{r,b}^{\mathrm{rep}}} - P_{0,r,b}\|_{2\to 2} \leq e^{-t\delta_*}. \tag{6}$$

Proof. Every $P_{v,r,b}$ is an orthogonal conditional expectation. The Dobrushin comparison applied to item (G3) gives (AT), so for $f \perp \mathbf{1}$,

$$\langle f, L_{r,b}^{\mathrm{rep}} f\rangle = \sum_v c_{v,r,b} \|(I - P_{v,r,b})f\|_2^2 \geq \frac{c_*}{A_*}\|f\|_2^2.$$

This proves the operator bound, and the spectral theorem gives the transfer bound. The four uniformities follow because items (G1)–(G4) use one type table and one approximate-tensorization modulus on the entire cofinal tower. □

Proposition 8.2 (Finite countermodels for the two hypotheses)**.** *If uniform mixing is removed, the faithful two-site family*

$$\pi_\varepsilon(00) = \pi_\varepsilon(11) = \frac{1-\varepsilon}{2}, \qquad \pi_\varepsilon(01) = \pi_\varepsilon(10) = \frac{\varepsilon}{2}$$

has exact heat-bath spectrum $\{0, 2\varepsilon, 2(1-\varepsilon), 2\}$*, hence gap* $2\varepsilon \to 0$*. If locality is removed, put the product fair-bit law on* $\{0,1\}^m$*, list its* 2^m *states in cyclic Gray-code order, and average over the two alternating perfect matchings, with projections* E_0, E_1*. Product mixing is exact and separated CMI is zero, but deciding which bit to change reads the whole configuration. Exact cycle Fourier modes give*

$$\operatorname{gap}\big((I - E_0) + (I - E_1)\big) = 1 - \cos(2\pi/2^m) \longrightarrow 0. \tag{7}$$

Thus both countermodels are finite, and neither hypothesis can be deleted.

Remark 8.3 (What the certificate must contain)*.* Finite-range Gibbs form and collar-CMI decay do not imply (3) and do not control projection angles. A numerical receipt must certify outward-rounded bounds $c_*^{\mathrm{lo}} > 0$ and $\eta_*^{\mathrm{hi}} < 1$, yielding the explicit certified floor $\delta_*^{\mathrm{lo}} = c_*^{\mathrm{lo}}(1 - \eta_*^{\mathrm{hi}})$. Point estimates do not qualify.

Remark 8.4 (Executable calibration boundary)*.* The exact-rational 244-type four-dimensional Ising calibration has $c_* = 1$, $\eta_* \le 1/2$, and hence $\delta_* \ge 1/2$ for its declared finite family. Its receipt is an executable test of the collar-type data model, not a physical compact-simple-gauge source receipt. A physical instance requires source collar kernels, refinement closure, gauge and zero-mode data, and independent continuum and transfer receipts. None of these data is supplied by the calibration.

9 Machine-Checked Boundaries and a Finite Gauge Diagnostic

Three finite results constrain how the receipts of Assumption 7.3 and Definition 7.5 can be discharged. Two are machine-checked theorems in the OPH Lean formalization [8]; the third is a finite numerical computation on two small interacting gauge systems, with an analytic free-limit control [9].

Noncommuting collars. The formalization contains the commuting-projection special case of Theorem 8.1, in which pairwise commuting collar projections give $L_r^{\mathrm{rep}} \ge c_*(I - P_0)$. In that legacy Lean theorem, `thm_7_3_finite_gap`, the strictly positive collar rates are direct hypotheses. The separately machine-checked legacy Lemma 7.2 proves a positive scalar for its own uniform-fiber matrix datum, but no Lean theorem bridges that matrix and scalar to the collar projections and rates of the commuting theorem. The module's minimal zero-projection witness has joint fixed space $\{0\}$; on its one-point coordinate type the constant functions are the whole space. It therefore witnesses only the stripped finite operator premises and supplies neither a compact-gauge realization nor a continuum or physical mass gap.

The same formalization also contains a three-state rational countermodel: two proper symmetric idempotent Markov projections that fail to commute, whose unit-rate generator has eigenvalues $1/2$ and $3/2$ in place of the commuting subset sums 0, 1, and 2. Spectral accounting by subset sums does not transfer to the noncommuting collars of Definition 7.5. The Dobrushin route in Theorem 8.1 is the route that applies, and commutation is used nowhere in it. This provenance boundary does not rule out that route, the fiber-dependent-rate route below, or another receipt-complete tower.

Uniform gap and carrier growth. A second machine-checked theorem states that a refinement interface carrying a uniform bound on carrier cardinality cannot also carry a cofinal family with strict carrier growth. An admissible collar tower has strictly growing carriers along refinement, so the uniform floor (4) must be stated through the finite type table, as in Proposition 7.6, and never through a bound on carrier size.

The finite transfer receipt on $\mathbb{Z}_2$ gauge theory. Assumption 7.3 can be evaluated by finite-matrix methods on a specified finite gauge system. The accompanying float64 eigendecomposition and matrix-logarithm computation evaluates it for $\mathbb{Z}_2$ lattice gauge theory on an $L \times L$ periodic spatial torus, $L \in \{2, 3\}$, in the Gauss-law sector, with one collar per spatial link and the π_r-preserving heat bath on the fiber $\{o, X_l o\}$, where X_l flips link l and o ranges over gauge orbits. The ground-state transform is the Doob transform by the Perron vector Ω, with $\pi_r = \Omega^2$. Two transfer objects are tested: the reflection-positive Wilson transfer matrix $T = e^{\beta_s P/2} K e^{\beta_s P/2}$, with $K(s, s') = \prod_l e^{\beta_t s_l s'_l}$, P the spatial plaquette sum, and $H = -\log(T/\lambda_{\max})$; and the Kogut–Susskind Hamiltonian $H = -\lambda \sum_l X_l - \sum_p U_p$.

For the Wilson transfer matrix the receipt holds exactly at $\beta_s = 0$, where $K = \prod_l (e^{\beta_t} I + e^{-\beta_t} X_l)$ is a constant multiple of $\exp\left(\frac{1}{2} \log\coth \beta_t \sum_l X_l\right)$ and every rate equals the dual coupling $\log\coth \beta_t$; the relative residual is below 10^{-12} for both tori. At every tested interacting point $\beta_s = \beta_t \in \{0.1, 0.3, 0.5, 0.7, 1\}$ it fails. In these finite examples $\log T$ couples configurations at multiple Hamming distances, and the best constant-rate approximant $\sum_l c_l (I - E_l)$ leaves a relative Frobenius residual of 0.038 at $\beta_s = \beta_t = 0.1$, 0.084 at 0.5, and 0.24 at 1.0 on the $L = 3$ torus.

For the Kogut–Susskind Hamiltonian the single-flip structure survives the transform exactly,

$$L^{\mathrm{rep}} = \sum_l c_l(o)\,(I - E_l), \qquad c_l(o) = \lambda\big(r_l(o) + r_l(o)^{-1}\big), \qquad r_l(o) = \frac{\Omega(o)}{\Omega(X_l o)},$$

so the scalar cross-fiber condition of Assumption 7.3 holds only where $|\log\Omega(o) - \log\Omega(X_l o)|$ has one fixed value under each flip of link l. The measured spread $\max_o c_l(o)/\min_o c_l(o)$ on the $L = 3$ torus is 1.036 at $\lambda = 2$, 1.17 at $\lambda = 1$, and 2.05 at $\lambda = 1/2$. The Dobrushin sum η_* of the single-link conditional kernels is below one, among the interacting test points, at $\beta_s = \beta_t = 0.1$, where $\eta_* = 0.59$, and at $\lambda = 2$, where $\eta_* = 0.82$; at the other tested interacting couplings item (G3) fails on this system. The free control is assessed separately.

The diagnostic fixes three facts on the two tested finite systems and coupling grid. It is exact in the free limit. The tested Wilson transfer matrices fail through multi-flip terms, and the tested local Hamiltonians fail the scalar cross-fiber clause through fiber-dependent rates. This finite grid does not exclude anisotropic couplings, alternate transfer objects, fiber-dependent positive rates, or another compact-gauge tower. A receipt on a physical compact-gauge tower may therefore admit fiber-dependent rates, replacing each scalar c_C by a positive fiber-constant function bounded below by c_*, or name a transfer object other than the Wilson matrix whose ground-state transform is a constant-rate heat-bath sum. The floor in Theorem 8.1 survives the first option without change, since its proof uses only $c_{v,r,b} \geq c_*$ and each rate function is constant on its own collar fiber; multiplication then commutes with the corresponding conditional expectation. In the tested Kogut–Susskind transform, $c_l = \lambda(r_l + r_l^{-1}) \geq 2\lambda$ by AM–GM. The variable-rate comparison is machine-checked at its finite level (`Lean/Screen/KogutSusskindFiberRateComparison.lean`): the single-fiber rate identity $c_l = \lambda(r + r^{-1})$ with the sharp bound $c_l \geq 2\lambda$ and equality exactly at $r = 1$, and, for a finite independent product of two-point fibers with fiber-dependent rates, a product-generator spectral gap equal to the minimum fiber rate with explicit eigenstructure, hence bounded below by 2λ uniformly in every fiber ratio. Concrete two- and three-fiber instances with distinct ratios carry computed exact gaps in a released receipt tied to the finite-transfer conventions of this section. The comparison is proved on the independent fiber product, and the finite descent-and-restriction part of the gauge-orbit quotient step is machine-checked (`Lean/Screen/GaugeOrbitQuotientGap.lean`): the $\mathbb{Z}_2$ XOR-mask action, which agrees with the gauge action of the committed non-self-loop $L = 2, 3$ transfer receipts, is formalized as a permutation subgroup whose every element is a mask translation; with fiber weights symmetric on every moved link, a condition proved equivalent to ratio one there under the $\pi = \Omega^2$ reading and exactly the equality case of the 2λ floor, the product generator commutes with every gauge translation and restricts to the gauge-invariant subspace, and every product-space Dirichlet constant survives that restriction, so the minimum fiber rate and the 2λ floor bound the quotient generator on mean-zero invariant observables, uniformly in the unmoved-link ratios. A worked abstract two-bit global-complement instance has exact invariant-subspace constant $c_0 + c_1 = 4$ at $\lambda = 1$, strictly above the floor 2, so the bound is one-sided. This instance is not a physical one-site periodic lattice: repeated self-loop incidences cancel over $\mathbb{Z}_2$, and the finite diagnostic is defined only for $L \geq 2$. The quotient is formalized as the invariant subspace; the orbit-space carrier formulation, the committed receipt tori with several star generators, the non-product orbit law with orbit-dependent rates, and the approximate-tensorization comparison are the named residual steps, and testing the anisotropic Wilson-to-Hamiltonian limit is the other live constructive route. The scalar form of Theorem 7.4 is what cross-fiber equality buys; the floor itself does not require scalar rates.

10 Continuum Extraction

Definition 10.1 (Finite compact-gauge state spaces and local cylinders)**.** *At regulator r, make the following construction both for the finite four-dimensional Euclidean slab $\Lambda_r^{(4)} = (V_r^{(4)}, E_r^{(4)})$ and for its time-zero spatial slice $\Lambda_r^{(0)} = (V_r^{(0)}, E_r^{(0)})$. In the formulas below $\Lambda_r = (V_r, E_r)$ denotes either choice until the superscript is restored. Let D_r be the finite repaired-record and zero-obstruction sector register. In the gauge-register presentation, let*

$$\widetilde{X}_r \subseteq G^{E_r} \times D_r$$

be the closed set obeying the finite Gauss, overlap, and repaired-record constraints. With $\mathcal{G}_r = G^{V_r}$ acting at edge endpoints and $\sim_{\mathrm{ov}}$ the closed relation of support-visible overlap/presentation indistinguishability, define

$$X_r := (\widetilde{X}_r/\mathcal{G}_r)/\sim_{\mathrm{ov}}$$

as the support-visible quotient. This is compact Hausdorff. The regulator has finitely many cells, but X_r need not be a finite set when G is a compact Lie group.

For each active collar C, let

$$\rho_{C,r} : X_r \to Y_{C,r}$$

be the continuous complete repaired readback. The complementary conditional fiber contains every record or holonomy coordinate actually resampled by repair and is either finite uniform or a standard atomless probability space with its conditioned MaxEnt/Haar law. The readback retains exactly the repaired fixed datum, not every pre-repair holonomy. A quantum-link carrier enters this proposition only through its commuting Euclidean cylinder/readout subalgebra; the noncommutative carrier algebra would require a separate GNS-representation-kernel support reduction.

For a bounded cell region $O \subset \Lambda_r$, let $\mathfrak{C}_r^{G,\mathrm{sv}}(O)$ be generated by the gauge-invariant Peter–Weyl matrix coefficients contracted into spin networks, Wilson-loop characters, and retained gauge-invariant boundary carriers in O, together with the overlap-sector projectors and continuous functions of the complete repaired collar readbacks in O. Define

$$\mathcal{A}_r^{G,\mathrm{sv}}(O) := \overline{\mathfrak{C}_r^{G,\mathrm{sv}}(O)}^{\|\cdot\|_\infty} \subseteq C(X_r).$$

Equivalently, start from the freely presented gauge-invariant readout-generator *-algebra, quotient by the kernel of its evaluation homomorphism on X_r, and complete. This kernel is a closed two-sided overlap-trivial ideal chosen before a state and is distinct from the limiting GNS null ideal below. The relation $\sim_{\mathrm{ov}}$ identifies exactly the configurations on which all declared spin-network, sector, and repaired-readback generators agree. The resulting self-adjoint unital algebra separates points of X_r, so Stone–Weierstrass gives

$$\mathcal{A}_r^{G,\mathrm{sv}}(\Lambda_r) = C(X_r).$$

For any selected stage measure π_r, its GNS cylinder closure is therefore the full $L^2(X_r, \pi_r)$, after the finite π_r-null support reduction.

Write the two resulting systems as

$$(X_r^{(4)}, \mathcal{A}_r^{G,\mathrm{sv},(4)}(O)) \quad \textit{and} \quad (X_r^{(0)}, \mathcal{A}_{r,0}^{G,\mathrm{sv}}(B)).$$

Euclidean-history restriction gives a continuous time-zero map $\tau_{0,r} : X_r^{(4)} \to X_r^{(0)}$.

For $r \preceq s$, coarse shadow multiplies the ordered fine-edge holonomies above each coarse edge and restricts the repaired records and sector labels. Gauge covariance and refinement associativity give

$$p_{sr} : X_s \to X_r, \qquad p_{tr} = p_{sr} \circ p_{ts},$$

with the four-dimensional/time-zero commuting square

$$\tau_{0,r} \circ p_{sr}^{(4)} = p_{sr}^{(0)} \circ \tau_{0,s},$$

and unital local *-maps on the separated support-visible quotient,

$$\iota_{rs}^O : \mathcal{A}_r^{G,\mathrm{sv}}(O) \to \mathcal{A}_s^{G,\mathrm{sv}}(O_s), \qquad \iota_{rs}^O(a) = a \circ p_{sr}, \qquad \iota_{rt}^O = \iota_{st}^{O_s} \iota_{rs}^O.$$

These maps preserve local isotony, gauge invariance, repaired readbacks, and the zero-obstruction sector. When this cylinder system is used on the Standard Model gauge paper's refinement-limit gauge branch, its sector-projector restriction must match the separately certified block-multiplicity injection in the compact-gauge refinement receipt [2]; that injection is not a consequence of deterministic coarse shadow. We use a countable cofinal regulator sequence and countable bounded-region exhaustion. The unrestricted directed version uses a subnet and product compactness instead. If every declared coarse configuration has a fine extension, p_{sr} is surjective and the full pullback is injective. Otherwise the C^*-inductive limit automatically quotients the common refinement-null kernel; Proposition 10.2 does not require prelimit injectivity. That noninjective quotient branch is not, by itself, the gauge paper's receipt-certified sector ladder. Write $\mathcal{A}_r^{G,\mathrm{sv}} := \mathcal{A}_r^{G,\mathrm{sv}}(\Lambda_r)$.

Proposition 10.2 (Projective weak-* extraction and support-visible GNS gluing)**.** *Let $\mu_s^{(4)}$ be the declared finite quotient-Gibbs probability measure on $X_s^{(4)}$, let*

$$\pi_s := (\tau_{0,s})_\# \mu_s^{(4)}, \qquad \mu_s^{(0)} := \pi_s,$$

and, for $d \in \{4, 0\}$, *define*

$$\omega_s^{(d)}(a) := \int_{X_s^{(d)}} a \, d\mu_s^{(d)}, \qquad \omega_{s\downarrow r}^{(d)} := \omega_s^{(d)} \circ \iota_{rs}^{(d)}.$$

The following statements hold for both values of d; the superscript is suppressed in the proof.

(i) *for* $q \preceq r \preceq s$,

$$\omega_{s\downarrow r} \circ \iota_{qr} = \omega_{s\downarrow q};$$

(ii) *a cofinal subsequence, or subnet, has weak-* limits* $\omega_{s_\alpha \downarrow r} \to \overline{\omega}_r$ *on every fixed local cylinder, and*

$$\overline{\omega}_r = \overline{\omega}_t \circ \iota_{rt} \qquad (r \preceq t);$$

(iii) *the local limits glue to a state* ω_∞ *on*

$$\mathcal{A}_\infty^{G,\mathrm{cyl}}(O) := \overline{\varinjlim_r \mathcal{A}_r^{G,\mathrm{sv}}(O_r)},$$

and the union of these isotone local algebras is dense in the global cylinder algebra;

(iv) *after the separate support reduction*

$$\mathcal{N}_\omega(O) := \{a \in \mathcal{A}_\infty^{G,\mathrm{cyl}}(O) : \omega_\infty(a^*a) = 0\}, \qquad \mathcal{A}_\infty^{G,\mathrm{supp}}(O) := \mathcal{A}_\infty^{G,\mathrm{cyl}}(O)/\mathcal{N}_\omega(O),$$

the induced state and representation are faithful. The compatible local GNS spaces have a Hilbert direct limit $(K^{(d)}, \Pi^{(d)}, \mathbf{1}^{(d)})$, *and the union of their cylinder images is dense. Write* $K^E := K^{(4)}$ *for the Euclidean-history GNS space and* $K := K^{(0)}$ *for the time-zero repair GNS space. The latter vector becomes the physical vacuum only through the transfer/OS certificate.*

If, in a finite quotient presentation, the four-dimensional measures have declared weights

$$\mu_r^{(4)}(x) = Z_r^{-1} m_r(x) e^{-S_r(x)}$$

and satisfy the fiber-sum identity

$$\sum_{x' : p_{sr}(x') = x} m_s(x') e^{-S_s(x')} = \alpha_{sr} m_r(x) e^{-S_r(x)} \tag{FS}$$

with α_{sr} *independent of* x, *then* $(p_{sr}^{(4)})_\# \mu_s^{(4)} = \mu_r^{(4)}$. *The time-zero square gives* $(p_{sr}^{(0)})_\# \pi_s = \pi_r$, *so both original state families are projective. Without (FS), compactness produces a projective cluster family; it does not prove* $\overline{\omega}_r = \omega_r$ *or uniqueness of the continuum state. For continuous compact-holonomy stages, replace (FS) by the corresponding disintegration receipt* $(p_{sr}^{(4)})_\# \mu_s^{(4)} = \mu_r^{(4)}$.

Proof. For $q \preceq r \preceq s$, coarse-shadow functoriality gives

$$(\omega_s \circ \iota_{rs}) \circ \iota_{qr} = \omega_s \circ \iota_{qs},$$

proving (i).

For a fixed local cylinder algebra, its state space is weak-* compact by Banach–Alaoglu. The chosen local algebras are separable because compact metrizable G has a countable Peter–Weyl test algebra and a finite regulator has only finitely many record and sector generators. Their state spaces are therefore weak-* metrizable. Successively extract a subsequence for the first, second, and later cylinders, choosing the k-th diagonal index beyond the k-th regulator stage. The diagonal subsequence is cofinal and converges on all cylinders. For a general directed regulator set, use a convergent subnet in the compact product of the local state spaces; unresolved early coordinates may be filled arbitrarily because each fixed coordinate is genuinely resolved on a cofinal tail. Weak-* continuity of precomposition and part (i) yield

$$\overline{\omega}_r \circ \iota_{qr} = \lim_{\alpha} \omega_{s_\alpha \downarrow r} \circ \iota_{qr} = \lim_{\alpha} \omega_{s_\alpha \downarrow q} = \overline{\omega}_q,$$

which proves (ii).

For a representative $[a]_r$ in the algebraic inductive limit, define

$$\omega_\infty([a]_r) := \overline{\omega}_r(a).$$

Compatibility makes this well defined. Positivity and normalization are checked in one finite algebra containing the element. For every $t \succeq r$,

$$|\overline{\omega}_r(a)| = |\overline{\omega}_t(\iota_{rt}(a))| \leq \|\iota_{rt}(a)\|.$$

Taking the infimum along the tail gives the inductive-limit norm bound even for noninjective bonding maps, so the state extends uniquely to the C^*-completion. This proves (iii).

On a commutative local cylinder algebra, the Riesz representation theorem identifies the null ideal with the continuous functions that vanish on the measure support. The support quotient is therefore $C(\operatorname{supp} \mu_O)$, where μ_O is the representing local Radon measure, and its induced state is faithful. If $O \subseteq O'$, state compatibility gives

$$a \in \mathcal{N}_\omega(O) \quad \Longleftrightarrow \quad \iota_{OO'}(a) \in \mathcal{N}_\omega(O').$$

Thus the inclusions descend injectively and the support quotients form an isotone faithful local net. For $r \preceq t$, define

$$V_{rt}[a]_r := [\iota_{rt}(a)]_t.$$

It is well defined and isometric because

$$\|V_{rt}[a]_r\|^2 = \overline{\omega}_t(\iota_{rt}(a^*a)) = \overline{\omega}_r(a^*a).$$

These maps compose, preserve $\mathbf{1}$, and intertwine the cylinder representations. Their Hilbert direct limit is the GNS completion of the cylinder union, proving (iv).

Finally, suppressing the $d = 4$ superscript, summing (FS) over x gives $Z_s = \alpha_{sr} Z_r$. Division by the partition functions then gives $(p_{sr}^{(4)})_\# \mu_s^{(4)} = \mu_r^{(4)}$, equivalently $\omega_s^{(4)} \circ \iota_{rs}^{(4)} = \omega_r^{(4)}$. The commuting time-zero square gives the corresponding $d = 0$ identity. □

Remark 10.3 (What compactness does not preserve). The overlap-trivial presentation quotient and the GNS support quotient are different. Even faithful finite states can converge to a nonfaithful state. Proposition 10.2 constructs the cylinder algebra, its state, the repair-side GNS space K, and the constant vacuum vector. It does not construct $\mathcal{H}$, a limit generator, a physical intertwiner, or vacuum-projection convergence. Those require the finite dynamic receipts below.

Assumption 10.4 (Four-dimensional Yang–Mills continuum certificate)**.** *In addition to the finite compact-gauge repair data, one cofinal regulator family supplies all of the following.*

(Y1) *For each bounded region O, renormalized gauge-invariant cylinder maps from one fixed countable test *-algebra $\mathfrak{C}_0^G(O)$ to $\mathcal{A}_r^{G,\mathrm{sv}}(O_r)$, with a uniform finite-volume/lattice-spacing Cauchy bound. The maps commute with the coarse-shadow ι_{rs} up to those vanishing Cauchy defects. This makes the extracted cluster family unique on the declared tests.*

(Y2) *Exact finite reflection positivity, or a positive self-adjoint transfer matrix, with reflection-compatible renormalization maps.*

(Y3) *Uniform restoration of Euclidean covariance, locality, regularity, clustering, and the declared vacuum-sector condition.*

(Y4) *Generalized Mosco convergence of both the physical transfer forms q_r^{phys} and repair forms q_r^{rep} to closed nonnegative continuum forms, or exact coherent direct-limit semigroup identities under state-preserving repair and physical refinement isometries, with one refinement-independent physical Euclidean-time normalization. In the exact alternative the isometries send $\mathbf{1}_r$ to $\mathbf{1}_s$ and Ω_r to Ω_s.*

(Y5) *Convergence of the finite intertwiners U_r and their inverses on dense form cores to a unitary U, together with $U_r\Omega_r = \mathbf{1}_r$ and strong convergence of $P_{0,r} = |\mathbf{1}_r\rangle\langle\mathbf{1}_r|$ to $P_0^K = |\mathbf{1}\rangle\langle\mathbf{1}|$ under the declared Hilbert identifications. The physical vacuum projection is $P_0^{\mathcal{H}} := U^{-1}P_0^K U = |\Omega\rangle\langle\Omega|$. On the dense time-zero cylinder core, the finite Markov/OS identity*

$$\omega_r^{(4)}\Big((\iota_{0,r}f)^*\tau_t^{(r)}(\iota_{0,r}g)\Big) = \langle f, e^{-tL_r^{\mathrm{rep}}}g\rangle_{K_r} = \langle U_r^{-1}f, e^{-tH_r}U_r^{-1}g\rangle_{\mathcal{H}_r}$$

holds and passes to the limit. Thus the OS-reconstructed time-zero Hilbert space and Hamiltonian are the same $(\mathcal{H}, H)$ as the transfer-form limit, not a second pair with reused notation.

(Y6) *A nontriviality certificate: renormalized local observables A_r, all images of one fixed bounded cylinder test under the maps in (Y1), whose centered vectors converge under the Hilbert identifications and satisfy*

$$\inf_r \mathrm{Var}_{\omega_r}(A_r) > 0, \qquad \sup_r q_r^{\mathrm{phys}}(A_r\Omega_r) < \infty.$$

The explicit multiresolution construction in the main paper supplies a model for items (Y1)–(Y2) and an exact transfer tower when its shell identities are implemented. Identification of its continuum limit with the four-dimensional Yang–Mills OS theory is the additional content of (Y3)–(Y6).

Theorem 10.5 (Conditional continuum transfer identification)**.** *Under Assumption 10.4, the generalized Mosco limit, or the exact coherent direct limit in item (Y4), produces nonnegative self-adjoint generators H and L^{rep} and a unitary U satisfying*

$$Ue^{-tH}U^{-1} = e^{-tL^{\mathrm{rep}}} \qquad (t \geq 0), \tag{8}$$

and hence

$$UHU^{-1} = L^{\mathrm{rep}}. \tag{9}$$

If the finite repair generators satisfy

$$L_r^{\mathrm{rep}} \geq \delta_*(I - P_{0,r})$$

with one $\delta_* > 0$ and the vacuum projections converge strongly, then

$$H \geq \delta_*(I - P_0^{\mathcal{H}}).$$

Proof. Generalized Mosco convergence is equivalent to generalized strong-resolvent and transfer-semigroup convergence on the declared Hilbert identifications. In the exact alternative, the coherent finite contraction semigroups define strongly continuous direct-limit semigroups on the dense cylinder images, and uniform contractivity extends them to the completed Hilbert spaces. Their bounded operators are symmetric and positive on the dense finite-stage union, hence self-adjoint and positive after extension. Convergence of U_r and U_r^{-1} on dense form cores passes the finite-stage intertwining relation to the limit, so uniqueness of self-adjoint semigroup generators gives $UHU^{-1} = L^{\mathrm{rep}}$. For a repair-form Mosco recovery sequence $\psi_r \to \psi$,

$$q_r^{\mathrm{rep}}(\psi_r) \geq \delta_* \big(\|\psi_r\|^2 - \|P_{0,r}\psi_r\|^2\big).$$

Taking the limit and using strong convergence of the vacuum projections gives the continuum form inequality. In the exact direct-limit alternative, the finite bound is equivalently

$$\|e^{-tL_r^{\mathrm{rep}}}(I - P_{0,r})\| \leq e^{-\delta_* t};$$

coherence passes this estimate to the direct-limit semigroup, and the spectral theorem gives the same continuum form inequality. □

11 Axiomatic Reconstruction and Nontriviality

Theorem 11.1 (Conditional Osterwalder–Schrader reconstruction on the support-visible compact–gauge branch)**.** *Under Assumptions 5.1 and 10.4, the continuum support-visible compact-gauge cylinder family is Euclidean invariant, reflection positive, regular on gauge-invariant local cylinder observables, nontrivial, and cyclic for the vacuum sector. Hence Osterwalder–Schrader reconstruction gives a four-dimensional quantum Yang–Mills theory*

$$(\mathcal{H}, \Omega, H, \mathcal{A}_{\mathrm{loc}}^G)$$

on the support-visible gauge-invariant local algebra, with $H \geq 0$ and e^{-tH} equal to the certified continuum Euclidean transfer semigroup.

Proof. Euclidean covariance, locality, reflection positivity, regularity, clustering, transfer convergence, and nontriviality are precisely the certificate items in Assumption 10.4. Proposition 10.2 supplies the four-dimensional Euclidean cylinder state K^E and its time-zero GNS restriction K; the certificate supplies the missing regularity and transfer controls. The finite/limiting time-zero identity in (Y5) identifies the OS time-zero inner product and translation semigroup with the transfer-form limit $(\mathcal{H}, H)$. The standard Osterwalder–Schrader reconstruction theorem therefore produces the Hilbert space, vacuum, local algebra, and positive Hamiltonian, and identifies the physical time-translation semigroup with the certified Euclidean transfer semigroup [13, 14]. □

Proposition 11.2 (Nontriviality of the support-visible compact-gauge theory)**.** *Under Assumption 10.4, the support-visible compact-gauge local algebra on the zero-obstruction vacuum branch strictly contains the vacuum scalars and admits a non-vacuum finite-energy local excitation.*

Proof. The compact-gauge witness and physical-UV landing theorem in the OPH Standard Model gauge paper supplies a realized nontrivial compact-gauge branch [2]. At finite cutoff this gives a support-visible gauge-invariant local observable, such as a nonconstant Wilson/plaquette cylinder observable, whose vacuum variance is positive. Its GNS vector is orthogonal to the vacuum after subtracting its expectation value. The finite-range local-Gibbs generator assigns finite energy to finite-cylinder excitations. The continuum nontriviality step is not a consequence of weak-$*$ state convergence alone; it is the variance-floor and finite-energy item (Y6) in Assumption 10.4. With that certificate, the local cylinder vectors survive in the support-visible continuum, so the continuum local algebra is strictly larger than $\mathbb{C}I$ and contains non-vacuum finite-energy local excitations. □

12 Main Theorem

Theorem 12.1 (Conditional positive four-dimensional compact-gauge Yang–Mills mass gap)**.** *Let G be a compact simple gauge group carried by a support-visible compact-gauge OPH vacuum branch satisfying the standing setup above and the continuum certificate of Assumption 10.4, the finite transfer receipt of Assumption 7.3, and all hypotheses of Theorem 7.4, Lemma 7.6, and Proposition 8.1. The theory reconstructed in Theorem 11.1 is nontrivial, and its continuum support-visible Hamiltonian H satisfies*

$$H \geq \delta_*(I - P_0^{\mathcal{H}}), \tag{10}$$

where $P_0^{\mathcal{H}}$ projects onto the physical vacuum. Therefore

$$\operatorname{Spec}(H) \cap (0, \delta_*) = \varnothing, \qquad \Delta_{\mathrm{YM}} \geq \delta_* > 0. \tag{11}$$

On that certified continuum branch the repair gap and Yang–Mills gap are exactly equal:

$$\Delta_{\mathrm{YM}} = \Delta_{\mathrm{rep}}. \tag{12}$$

Proof. At every finite stage, Proposition 8.1 gives

$$L_r^{\mathrm{rep}} \geq \delta_*(I - P_{0,r}).$$

By Theorem 10.5, on the extracted cylinder/GNS space, this lower bound passes to the support-visible continuum:

$$L^{\mathrm{rep}} \geq \delta_*(I - P_0^K).$$

Using (9), $UHU^{-1} = L^{\mathrm{rep}}$. Conjugating the lower bound by U^{-1} gives (10), and the spectral statement (11) follows. Since unitary equivalence preserves the nonzero spectrum and identifies the vacuum with the constant sector, the infimum of the nonzero spectrum is the same on both sides, giving (12). □

Remark 12.2 (Group-uniform form). The proof is group-uniform. Once a compact simple G is carried by a compact-gauge zero-obstruction OPH branch, no step uses special properties of the realized Standard Model quotient. The inputs are compact-gauge support-visible quotient locality, exact local repair on collars, the finite ground-state-transform/cross-fiber receipt, the source-type and uniform approximate-tensorization receipts, refinement coherence, projective cylinder extraction, the transfer/vacuum certificate, and the OS regularity/noncollapse certificate.

13 Exact Gap Accounting

On a branch satisfying Assumption 10.4, the proof gives more than a positive lower bound. It identifies the Hamiltonian whose gap is being measured:

$$H = U^{-1} L^{\mathrm{rep}} U.$$

Therefore

$$\mathrm{Spec}(H) \setminus \{0\} = \mathrm{Spec}(L^{\mathrm{rep}}) \setminus \{0\}.$$

The Yang–Mills gap is exactly the first nonzero repair eigenvalue:

$$\Delta_{\mathrm{YM}} := \inf\big(\mathrm{Spec}(H) \setminus \{0\}\big) = \inf\big(\mathrm{Spec}(L^{\mathrm{rep}}) \setminus \{0\}\big) =: \Delta_{\mathrm{rep}}.$$

The finite-stage approximate-tensorization argument proves $\Delta_{\mathrm{rep}} \geq \delta_* > 0$. The exact accounting statement is the conditional equality $\Delta_{\mathrm{YM}} = \Delta_{\mathrm{rep}}$; the inequality is the positivity proof for that same quantity. Without the continuum certificate, the same equations are finite-regulator repair accounting instead of a Clay-admissible Yang–Mills theorem.

14 Relation to the 2D Yang–Mills Bridge

The companion OPH results also contain an exact edge-sector identity:

$$Z_{\mathrm{edge}}(t) = \sum_R d_R^2 e^{-tC_2(R)} = K_t(1),$$

which is the compact-group heat kernel at the identity and matches the standard two-dimensional Yang–Mills heat-kernel partition form. Peter–Weyl supplies the heat-kernel identity, and Gross–Taylor gives the standard large-N worldsheet dictionary when a separate large-N_{edge} branch with remainder control is declared [15, 16].

That 2D result is a normalization and worldsheet-effective bridge. The spectral theorem above concerns the support-visible compact-gauge Hamiltonian and obtains its lower bound by identifying Euclidean transfer with the repair generator. The string-vacuum selector uses the same bridge only as effective edge-language input; its critical-worldsheet, Bouchard-Donagi, safety-layer, threshold, and moduli-locking conditions do not alter the four-dimensional compact-gauge mass-gap theorem surface [7].

15 Requirements for a Clay-Level Construction

Clay/Jaffe–Witten target	OPH repair-dynamics boundary
Compact simple gauge group G	G is arbitrary compact simple, provided it is carried by the OPH compact-gauge zero-obstruction branch.

Clay/Jaffe–Witten target	OPH repair-dynamics boundary
Four-dimensional Euclidean Yang–Mills form	Theorem 5.2, imported from The Standard Model gauge paper's Yang–Mills theorem surface, gives $S_E[A] = \frac{1}{4g^2}\int_{\mathbb{R}^4}\langle F_{\mu\nu}, F_{\mu\nu}\rangle\, d^4x$ under the branch and renormalized identification certificates.
Four-dimensional quantum Yang–Mills theory	Proposition 10.2 proves support-visible projective weak-* / GNS extraction. Renormalized Schwinger convergence, reflection positivity, Euclidean covariance/locality, clustering, nontriviality, and transfer/intertwiner convergence are conditional on Assumption 10.4.
Nontriviality	Conditional on the variance-floor and finite-energy certificate in item (Y6). Finite nonconstant Wilson/plaquette cylinders are candidates; weak-$*$ convergence alone is not enough.
Axiomatic strength	Theorem 11.1 states the conditional OS reconstruction step on the support-visible gauge-invariant local algebra.
Mass gap	Under the finite ground-state-transform, source-type, cross-fiber-rate, and uniform approximate-tensorization receipts, $L_r^{\mathrm{rep}} \geq \delta_*(I - P_{0,r})$, where $\delta_* = c_*/A_*$. The continuum Hamiltonian bound $H \geq \delta_*(I - P_0^{\mathcal{H}})$ follows only after Theorem 10.5.
Exact gap accounting	On the certified continuum branch, $UHU^{-1} = L^{\mathrm{rep}}$ gives $\Delta_{\mathrm{YM}} = \Delta_{\mathrm{rep}}$. OPH accounts exactly for the Yang–Mills gap as the first nonzero repair eigenvalue only on that branch.
Uniformity in G	The proof uses compact simplicity and compact-gauge quotient locality, not Standard-Model-specific representation data.

16 Conclusion

The OPH finite repair mechanism for the Yang–Mills mass gap is simple. A non-vacuum support-visible compact-gauge excitation is not fixed by all local repair collars. At least one active collar must relax it. Because active collar rates have a uniform positive floor and the source law has one uniform approximate-tensorization constant, every non-vacuum finite-regulator state pays a positive Euclidean repair cost. The finite-stage statements are exact theorems on the declared cylinder system, with every assumption named, the load-bearing receipt isolated in one place, and that receipt evaluated on a finite gauge system; the mechanism inherits the wider program's discipline of finite core theorems and zero continuous dials.

The Clay-facing conclusion is conditional. If the regulator family also satisfies the renormalized continuum, reflection-positivity, OS, nontriviality, and transfer/intertwiner certificate of Assumption 10.4, then the continuum compact-gauge Hamiltonian is unitarily equivalent to the repair generator. On that certified branch the Yang–Mills gap is the repair gap:

$$\boxed{\Delta_{\mathrm{YM}} = \Delta_{\mathrm{rep}} \geq \delta_* = c_*/A_* > 0.}$$

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